



Real-Time Ferry Ticket Sales and Redemption Analytics for Toronto Island Park Using Power BI.

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ABSTRACT

This project focuses on analysing real-time ferry ticket sales and redemption data for Toronto Island Park using Power BI and DAX analytics. The study aims to identify passenger movement trends, peak demand periods, and operational congestion patterns. Historical and near real-time data from 2015–2025 was analysed using data visualization techniques, KPI monitoring, and time-series analytics. The dashboard provides operational insights that can support ferry scheduling, improve passenger flow management, and enable data-driven decision-making.

INTRODUCTION

About Ferry Transportation

- Ferries transport thousands of passengers daily
- Peak-hour congestion affects efficiency
- Real-time monitoring improves planning

About Project

This project studies: -

- Total tickets sold by month
- Total tickets redeemed by month
- Total tickets redeemed and total tickets sold by hour
- Average tickets sold by day name
- Total tickets sold and net passenger movement by month
- Total tickets sold by isweekend
- Redemption rate % by gauge min and gauge max

OBJECTIVE

To analyse Real-Time Ferry Ticket Sales and Redemption for Toronto Island Park using data visualization and business intelligence tools.

PROBLEM STATEMENT

Although ferry ticket sales and redemptions are recorded every 15 minutes, there is no centralized analytics platform available to monitor passenger movement, identify peak demand windows, or support operational planning in real time.

OBJECTIVES

Primary Objectives

- Analyse historical and near real-time ticket activity
- Identify peak and off-peak demand periods
- Quantify passenger inflow and outflow trends

Secondary Objectives

- Support ferry scheduling decisions
- Improve passenger experience
- Reduce terminal congestion
- Enable operational planning

TOOLS & TECHNOLOGIES

Tool	Purpose
Microsoft Power BI	Dashboard & Visualization
DAX	KPI Calculations
Excel/CSV	Dataset Handling and Source
Power Query	Data Cleaning

DATASET DESCRIPTION

Column	Description
_id	Unique identifier
Timestamp	Date and time
Sales Count	Tickets sold
Redemption Count	Tickets redeemed

Dataset period: 2015–2025

Data interval: 15 minutes

METHODOLOGY

Data Ingestion

- CSV dataset imported into Power BI
- Timestamp converted into datetime format
- Data sorted chronologically

Data Cleaning

- Missing value handling
- Timestamp consistency checks
- Data validation

Feature Engineering

Created:

- Hour
- Month
- Year
- Day Name
- Weekend/Weekday indicators

Exploratory Data Analysis

Performed:

- Hourly trend analysis
- Daily demand analysis
- Peak demand identification
- Sales vs redemption comparison

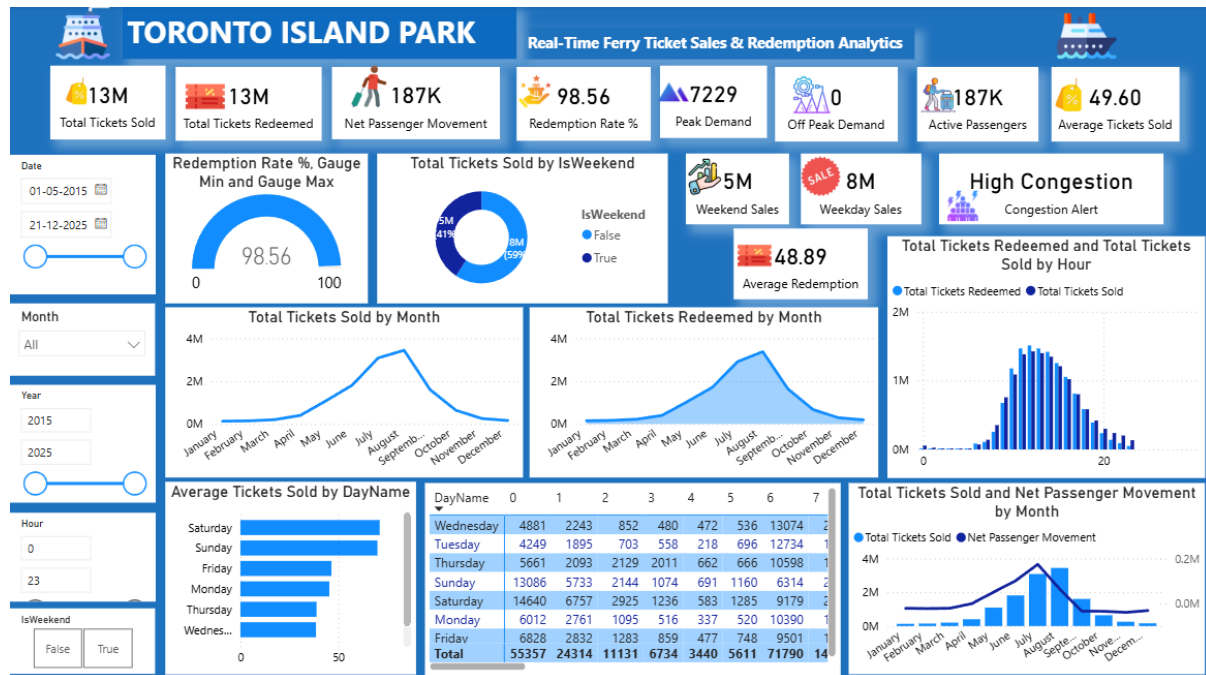
KPIS USED

- Total Tickets Sold
- Total Tickets Redeemed
- Net Passenger Movement
- Redemption Rate %
- Peak Demand
- Off Peak Demand
- Active Passengers
- Average Tickets Sold
- Average Redemption
- Weekend Sales
- Weekday Sales
- Congestion Alert

VISUALISATION USED

KPI	Description
Line Chart	Sales Trends
Area Chart	Redemption Trends
Donut Chart	Weekend vs weekday analysis
Gauge Chart	Redemption efficiency
Heatmap	Peak demand windows
Bar Chart	Daily Comparison

DASHBOARD OVERVIEW



KPI cards

- Total Tickets Sold – Shows the total number of ferry tickets sold during the selected period.
- Total Tickets Redeemed – Represents the total number of tickets scanned or redeemed at ferry terminals.
- Net Passenger Movement – Displays the difference between tickets sold and redeemed to track passenger flow.
- Redemption Rate % – Measures the percentage of sold tickets that were successfully redeemed.
- Peak Demand – Identifies the highest passenger demand interval based on ticket sales.
- Off Peak Demand – Shows the lowest passenger demand period.
- Active Passengers – Represents the estimated number of passengers currently active in the system.
- Average Tickets Sold – Displays the average number of tickets sold over time.
- Average Redemption – Shows the average number of redeemed tickets.
- Weekend Sales – Measures total ticket sales during weekends.

- Weekday Sales – Represents total ticket sales during weekdays.
- Congestion Alert – Indicates whether passenger movement exceeds the congestion threshold.

Charts

- Line Chart (Ticket Sales Trend) – Displays ticket sales trends over time and helps identify peak demand periods.
- Area Chart (Redemption Trend) – Represents ticket redemption activity and passenger boarding patterns over time.
- Clustered Column Chart (Sales vs Redemption) – Compares tickets sold and redeemed to analyse passenger inflow and outflow.
- Donut Chart (Weekend vs Weekday Sales) – Shows the distribution of ticket sales between weekends and weekdays.
- Heatmap / Matrix (Peak Demand Analysis) – Identifies busiest hours and days using colour-based demand intensity analysis.
- Line and Clustered Column Chart (Net Passenger Movement) – Visualizes passenger accumulation and movement trends over time.
- Gauge Chart (Redemption Rate) – Displays operational efficiency based on ticket redemption percentage.
- Bar Chart (Average Demand by Day) – Compares average passenger demand across different weekdays.

Slicers and Filters

- Date Slicer – Allows users to filter dashboard data based on a selected date range.
- Month Slicer – Helps analyse monthly passenger trends and ticket activity.
- Year Slicer – Enables yearly comparison of ferry operations and demand patterns.
- Hour Slicer – Filters data based on specific hourly intervals for peak-hour analysis.
- Weekend Filter (IsWeekend) – Separates weekend and weekday passenger activity for comparison.
- Day Name Slicer – Allows filtering data by specific weekdays to analyse daily demand trends.

KEY FINDING & INSIGHTS

The analysis identified significant passenger demand during weekends and holiday periods. Peak ticket sales were observed during afternoon hours, while redemption activity closely followed sales trends. The dashboard also highlighted operational congestion during high-demand periods, indicating the need for improved ferry scheduling and resource allocation.

- Weekend demand higher
- Afternoon traffic peak
- Congestion windows identified
- Passenger movement patterns visible

EXPECTED IMPACT AND CHALLENGES FACED

Expected Impact

- Improved ferry scheduling
- Reduced congestion
- Better staffing allocation
- Data-driven decision-making
- Improved passenger experience

Challenges Faced

- Handling time-series data
- Real-time dashboard optimization
- Data consistency checks
- Visual layout balancing

CONCLUSION

The Real-Time Ferry Ticket Sales and Redemption Analytics dashboard successfully demonstrate how business intelligence tools can support transportation management and operational planning. By integrating KPI monitoring, trend analysis, and interactive visualizations, the system provides actionable insights that can improve ferry operations, passenger management, and decision-making efficiency.

FUTURE SCOPE

- Live API integration
- AI-based demand prediction
- Weather integration
- Mobile dashboard
- Real-time alerts

REFERENCES

- [Toronto Open Data Portal](#)
- [Microsoft Power BI Documentation](#)
- [DAX Function Reference](#)
- [Transportation Analytics Research Papers](#)